Reg No.									

SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM – 1(SET-A)

Programme: B.E./B.Tech (CSE/IT) | Max. Marks: 50
Course Code: SIIBLH21 | Duration: 2 Hours
Course Name: PROGRAMMING IN PYTHON | Date : 27.02.2025
Batch: (2024-2028) | Sem : II

Answer All (5 × 10 = 50)

Q. No.	Questions	CO
1.	The player starts at the entrance of an isolated village. They have three choices: 1. Go into the village. 2. Walk along the streets. 3. Return to the hometown. Based on their choice, different scenarios unfold: 1. If they go into the village, they encounter a wild dog. They can either fight, sneak past or run away. 2. If they walk along the street, they find a school. They can either open it or ignore it. 3. If they return to the hometown, they meet a wise old man who gives them a hint about the village.	
	or	
2	Write a Python function calculate_area that takes three arguments: shape (a string), length, and width. Return length * width for a rectangle, 0.5 * length * width for a triangle, and 3.14 * (length ** 2) for a circle (using length as the radius). If the shape is invalid, return -1.	
3.	Paresh has purchased a product from a shop for an EMI. With the help of Python language, develop a EMI Calculator to calculate the gross amount.	
	or	
4	Create a Library Management System where users can borrow and return books. The system asks for the number of books to	

	borrow, collects titles in a loop, and displays the borrowed list. Users can return books, which are removed if found; otherwise, a message is shown. The system runs in a loop until the user chooses to exit.	

5.	Paul has bought 3 tiers for his vehicle. He wants to find the smallest one. Help him calculate the area and circumference of the circle using Python Language.	
	or	
6	Use the concept of inheritance and extend a Vehicle class to create a Bike class with additional properties like fuel type and methods to display information. 415	

7.	Implement a product class to store product details like name, price, and stock quantity, and allows users to buy products and check stock levels. 404	
	or	
8	Sameer has bought a new car and wants t know the specifications of it. Develop the concept of parameterized constructor and display the output using python programming.	

9.	Using the concept of Polymorphism, develop an application that can support different types of animals such as dog, cat and cow, each making unique sounds with the help of python programming.	
	or	
10	A database analysis is required by doctors for analyzing the data of patients. Use python programming for creating the same using class patient.	